DG INTERNS HUB — TECHNICAL NOTES

TOPIC 03: SPLUNK VS WAZUH SIEM COMPARISON

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TABLE OF CONTENTS

1. Executive Overview & Definitions
2. Detailed Platform Comparison Matrix
3. Core Strengths & Limitations
4. Architectural & Query Language Differences
5. Practical SOC Scenario: Platform Selection & Triage
6. Conclusion & Analyst Recommendation

1. Executive Overview & Definitions

Selecting a SIEM solution depends on organizational budget, scalability requirements, and team expertise. **Splunk** is a proprietary, enterprise-grade data analytics and SIEM platform known for its flexibility and Search Processing Language (SPL). **Wazuh** is an open-source platform integrating SIEM, XDR, and File Integrity Monitoring (FIM), natively built on top of OpenSearch/Elasticsearch.

2. Detailed Platform Comparison Matrix

Comparison Factor	Splunk Enterprise	Wazuh Open Source
Licensing Model	Commercial (Ingestion volume or compute-based)	Free & Open Source (Apache License 2.0)
Query Language	Search Processing Language (SPL)	Lucene Query Syntax / KQL / OpenSearch PPL
Endpoint Agent	Splunk Universal Forwarder (Log streaming only)	Wazuh Agent (Log collection + Active Response + FIM)
Out-of-the-Box Rules	Splunk Security Essentials (Add-on required)	Includes pre-mapped MITRE ATT&CK rulesets
Storage Backend	Proprietary flat file index buckets	Elasticsearch / OpenSearch cluster

3. Practical SOC Scenario: Endpoint File Modification Triage

> **Scenario: Unauthorized Modification of Web Server Config**
>
> **Incident Context:** An unauthorized modification occurred in `/etc/nginx/nginx.conf` on Web Server 01.
>
> **Wazuh Detection:** Wazuh's native FIM module immediately detects the checksum change, logging Syscheck Rule 550 and triggering an alert natively without extra plugins.
>
> **Splunk Detection:** Requires Auditd or Sysmon logs ingested via Universal Forwarder, queried using SPL: `index=linux_audit file="/etc/nginx/nginx.conf" action=modified`.
>
> **SOC Conclusion:** Wazuh provides superior out-of-the-box endpoint FIM, while Splunk offers deeper contextual correlation across heterogeneous network sources.

4. Conclusion & Recommendation

Splunk is ideal for large-scale enterprise SOCs needing unlimited data flexibility and complex correlations. Wazuh is an outstanding cost-effective choice for SMB SOCs and organizations prioritizing built-in endpoint threat detection and compliance compliance straight out of the box.